

PAPER

Research on Flux-Weakening Control with Single Current Regulator of PMSM Based on Parameter Identification

Yihao WANG[†], Jianguo XI^{†a)}, and Chengwei XIE[†], *Nonmembers*

SUMMARY In order to reduce the fluctuation range of output speed and torque of permanent magnet synchronous motor at high speed, this paper proposes an improved flux-weakening control method of single current regulator of permanent magnet synchronous motor. Based on chaotic mapping and Monarch Butterfly Optimization, an adaptive chaos monarch butterfly optimization is proposed to use for online parameter identification. This algorithm generates a chaotic sequence distributed according to a specific pattern through a single chaotic mapping, and combines it with the results of the previous parameter identification to generate an initialized population. Secondly, the impact mechanism of different load curve positions on the smoothness and efficiency of motor operation was analyzed, and the influence of motor parameters on voltage limited elliptical distortion was analyzed. Finally, a method of giving the cross-axis voltage considering motor parameters is designed. The simulation and experimental results show that the parameters of the motor can be accurately identified online based on ACMBO. When the load is 85 N·m, the range of motor torque output error is effectively reduced.

key words: permanent magnet synchronous motor (PMSM), flux-weakening control, single current regulator, online parameter identification, ACMBO

1. Introduction

PMSM is widely used as a driving motor for electric vehicles due to its simple structure and reliable operation. In the driving conditions of electric vehicles, it is often necessary to drive the motor to operate at high speeds [1]–[4]. However, during high-speed operation of the motor, the parameters of the motor will constantly change with factors such as temperature, magnetic saturation effect, and current, which will seriously affect the control performance of the motor operating in weak magnetic areas. In weak field control, single current regulator reduces the magnetic field of the motor by adjusting the current, in order to maintain output power or torque during high-speed operation. The control system of the single current regulator weak field control strategy is simple and easy to implement deep weak field, but different cross axis voltage settings also affect the motor's speed and torque output capabilities [5], [6]. Accurate parameter identification is crucial. Firstly, appropriate parameters can ensure that the current regulator matches the characteristics of the motor system, achieving precise current control and ensuring the stability and accuracy of weak magnetic control. Secondly, if the parameter identification is not accurate, it

may lead to abnormal operation of the system in weak magnetic regions, such as increased torque ripple and decreased efficiency. Furthermore, accurate parameters help optimize the dynamic performance of the system, reduce response time, and improve the adaptability of the motor under different operating conditions. Therefore, weak magnetic control based on real-time online identification of motor parameters has important practical research significance.

The current mainstream parameter identification algorithms include recursive least squares, model reference adaptive, and intelligent algorithms [7]–[11], while optimization control algorithms include sliding mode control, SVPWM, fuzzy PI control, symbiotic biological search algorithm and model predictive control [12]–[17]. For example, Ref. [18] proposed a recursive least squares method based on fuzzy forgetting factor to identify the parameters of built-in permanent magnet synchronous motors, which improved the identification performance of RLS algorithm. Reference [19] introduced a first-order digital low-pass filter to identify motor parameters and improve the accuracy of parameter estimation. The cross-axis voltage setting method of weak magnetic control with a single current regulator is mainly aimed at improving the operating efficiency and stability of the motor [20]. Reference [21] improved the feedback loop and variable axis voltage setting method of the weak magnetic control system, added amplitude limiting processing, and enhanced the stability of the motor at high speeds. However, the above methods ignore the influence of motor parameters on the given cross axis voltage in weak magnetic control.

Therefore, this article proposes Adaptive Chaos Monarch Butterfly Optimization for online identification of motor parameters based on the variation law of PMSM parameters, which solves the inherent problem of weak global search ability and susceptibility to local optima in the algorithm; At the same time, based on the mathematical model of PMSM, the influence mechanism of motor parameters on voltage limit elliptical distortion was analyzed, and an improved single current regulator weak magnetic control method for cross axis voltage setting was proposed.

2. PMSM Mathematical Modeling

To simplify the analysis, a three-phase built-in permanent magnet synchronous motor (IPMSM) is assumed to be an ideal motor [25]. The voltage equation of IPMSM in the d-q axis rotating coordinate system can be expressed as:

Manuscript received August 18, 2024.

Manuscript revised November 14, 2024.

Manuscript publicized January 27, 2025.

[†]Henan University of Science and Technology, School of Vehicle and Transportation Engineering, Luoyang, 471003, China.

a) E-mail: 17761673081@163.com

DOI: 10.1587/transele.2024ECP5054

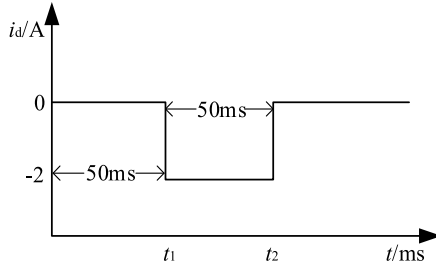


Fig. 1 Data sampling diagram

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \psi_f \end{cases} \quad (1)$$

In Eq. (1), u_d and u_q denote the components of the stator voltage on axes d and q; i_d and i_q represent the components of the stator current on axis d and axis q, respectively. L_d and L_q represent the inductance of axis d and axis q, respectively. R_s denotes the stator resistance of the motor; ψ_f is the permanent magnet flux; ω_e is the electric angular velocity of the rotor.

When the motor runs stably, it can be approximately considered that i_d and i_q do not change. By discretization of Eq. (1), the steady-state voltage equation can be obtained as follows:

$$\begin{cases} u_d(k) = R_s i_d(k) - \omega_e(k) L_q i_q(k) \\ u_q(k) = R_s i_q(k) + \omega_e(k) L_d i_d(k) + \omega_e(k) \psi_f \end{cases} \quad (2)$$

For the convenience of control, PMSM often adopts the control strategy of $i_d = 0$, simplifying Eq. (2) as follows:

$$\begin{cases} u_d(k) = -\omega_e(k) L_q i_q(k) \\ u_q(k) = R_s i_q(k) + \omega_e(k) \psi_f \end{cases} \quad (3)$$

To establish a full rank equation system, a negative sequence weak magnetic current can be injected into the d-axis, with a sampling period of 100 ms and data sampling conducted every 50 ms. Within one sampling period, two sets of data in states $i_d = 0$ and $i_d = -2$ A can be collected, as shown in Fig. 1.

Sending the collected data of one cycle to the upper computer can establish a full rank equation system under two states $i_d = 0$ and $i_d \neq 0$ as follows:

$$\begin{cases} u_{d0}(k) = -\omega_e(k) L_q i_{q0}(k) \\ u_{q0}(k) = R_s i_{q0}(k) + \omega_e(k) \psi_f \\ u_{d1}(k) = R_s i_{d1}(k) - \omega_e(k) L_q i_{q1}(k) \\ u_{q1}(k) = R_s i_{q1}(k) + \omega_e(k) (L_d i_{d1}(k) + \psi_f) \end{cases} \quad (4)$$

In Eq. (4), $u_{d0}(k)$, $u_{q0}(k)$, $i_{q0}(k)$ represents the k -th sampling data within time $0 - t_1$; $u_{d1}(k)$, $u_{q1}(k)$, $i_{d1}(k)$, $i_{q1}(k)$ represents the k -th sampling data within time $t_1 - t_2$. Each additional k requires a sampling time of 100 ms.

3. Adaptive Chaos Monarch Butterfly Optimization

3.1 Monarch Butterfly Optimization

Monarch Butterfly Optimization (MBO) is a population intelligent evolutionary algorithm that simulates the behavior of Monarch butterfly [22]. MBO divides all individuals in the population into two subpopulations P_1 and P_2 . Assuming the population size is n , the number of individuals in the two subpopulations is n_1 and n_2 respectively, and the proportion of the number of individuals in P_1 to the total population is p , then.

$$\begin{cases} n_1 = \text{ceil}(n * p) \\ n_2 = n - n_1 \end{cases} \quad (5)$$

In Eq. (5), ceil represents an upward rounding function.

The evolution of populations is based on two ways: migration and environmental adaptation. P_1 uses the migration method to update individual positions, as follows:

$$x_i^{t+1} = \begin{cases} x_{r_1}^t, \text{peri} * \text{rand} \leq p \\ x_{r_2}^t, \text{peri} * \text{rand} > p \end{cases} \quad (6)$$

In Eq. (6), t is the current number of iterations; x_i^{t+1} represents the position of the i -th particle in Generation $t + 1$; $x_{r_2}^t$ and $x_{r_1}^t$ represent randomly selecting individuals from subpopulations P_1 and P_2 ; peri is the migration cycle, usually taken as 1.2; rand represents a random number on $[0, 1]$.

Subpopulation P_2 updates individual positions through environmental adaptation, as follows:

$$x_i^{t+1} = \begin{cases} x_{best}^t, \text{rand} \leq p \\ x_{r_3}^t, \text{rand} > p \end{cases} \quad (7)$$

In Eq. (7), x_{best}^t represents the optimal individual in the population; $x_{r_3}^t$ represents a random individual selected from subpopulation P_2 .

Furthermore, if $\text{rand} > \text{BAR}$, then:

$$x_i^{t+1} = x_i^{t+1} + \frac{S_{\max}}{t^2} * (d_x - 0.5) \quad (8)$$

In Eq. (8), BAR represents the adjustment rate; $S_{\max}$ represents the maximum step size that can be moved in each step; d_x is the moving step size, which can be obtained through the *Levy* function.

$$d_x = \text{Levy}(x_i^t) \quad (9)$$

3.2 Population Initialization Based on Singer Chaotic Mapping and Information Transmission

In Monarch Butterfly Optimization, the initial population is generally generated randomly, and the uncertainty of individual distribution is high, which can easily lead to unstable identification results. The chaotic sequence generated

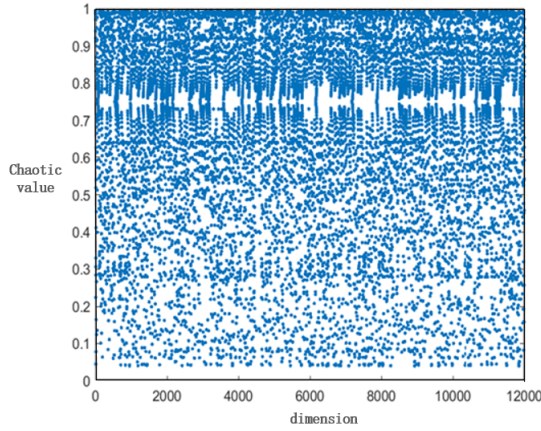


Fig. 2 Singer chaos sequence diagram

based on chaotic mapping has the characteristics of traversal, overall stability, and local instability. This article uses Singer mapping to generate chaotic sequences. When μ is set to 1.065, the chaotic sequence generated by this mapping has a decreasing number of particles according to the chaotic value. The number of individuals with chaotic values around 1 is the highest, and the number of individuals near 0 is the lowest. The population initialization algorithm steps for generating chaotic sequences using Singer mapping are as follows:

1. Set the population size to N , representing the number of individual monarch butterflies; The search dimension is D , representing the dimension of the position vector of each individual monarch butterfly.
2. For the first dimension ($j = 1$) of the first individual ($i = 1$), randomly generate a chaotic variable with an initial value $z_{1,1}$ in the $(0, 1)$ interval. Using the singer chaotic mapping equation shown in Eq. (10), iteratively generate chaotic variables for the other dimensions ($j = 2, \dots, D$) of the first emperor butterfly individual. Here, $\mu = 1.065$, this equation can generate chaotic behavior. The chaotic sequence generated by Eq. (10) is shown in Fig. 2:

$$z_{i,j+1} = \mu(7.86z_{i,j} - 23.31z_{i,j}^2 + 28.75z_{i,j}^3 - 13.302875z_{i,j}^4) \quad (10)$$

3. For the remaining $N - 1$ monarch butterfly individuals, repeat step 2 to randomly generate an initial chaotic variable in the $(0, 1)$ interval for the first dimension of each individual, and then use the singer chaotic mapping equation to generate chaotic variables for the other dimensions of the individual.

4. Map the chaotic variables based on the range of values of the individual position of the monarch butterfly in the MBO algorithm, and set the range of values for the j -th dimension of the individual position of the monarch butterfly $[a_j, b_j]$. Therefore, the actual value of the j -th dimension of the i -th individual position of the monarch butterfly is $y_{i,j} = a_j + z_{i,j}(b_j - a_j)$.

After the above steps, an initialized population was obtained where the positions of each individual monarch but-

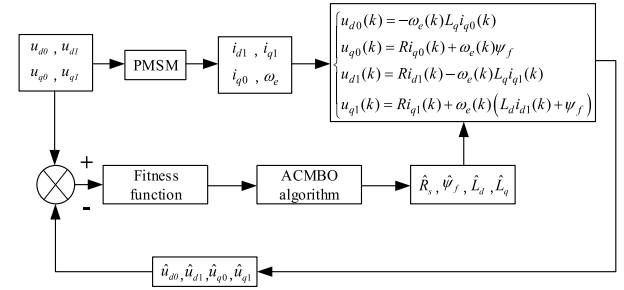


Fig. 3 Parameter identification model based on the ACMBO

terfly were within the corresponding solution space range.

For PMSM, it can be considered that the parameter changes are not significant in a short period of time without drastic changes in operating conditions. The identification results of adjacent parameters are similar, and the previous identification result can be used to provide guidance for the next parameter identification [23]. However, relying solely on one individual for information transmission can easily lead to information loss and fail to fully utilize the guiding role of the previous parameter identification results.

This article proposes to combine the previous parameter identification results with Singer chaotic mapping to generate a chaotic sequence distributed according to a specific pattern as the initialization population. Compared with the randomly generated initialization population, the number of individuals at different positions in the initialization population generated by this method is positively correlated with the previous identification results, with more individuals at positions closer to the previous identification results and fewer at positions farther away.

3.3 Specific Process of Parameter Identification

For a system with a known model but unknown parameters, the parameter identification problem can be transformed into a problem of finding the optimal value [24]. The principle of parameter identification is shown in Fig. 3. Among them, the fitness function can be defined as Eq. (11).

$$f = \sum_{k=1}^n \left[a_1 (u_{d0}(k) - \hat{u}_{d0}(k))^2 + a_2 (u_{d1}(k) - \hat{u}_{d1}(k))^2 + a_3 (u_{q0}(k) - \hat{u}_{q0}(k))^2 + a_4 (u_{q1}(k) - \hat{u}_{q1}(k))^2 \right] \quad (11)$$

In Eq. (11), a_1, a_2, a_3, a_4 represent the weight coefficients; $\hat{u}_{d0}(k), \hat{u}_{d1}(k), \hat{u}_{q0}(k), \hat{u}_{q1}(k)$ represent the calculated voltage.

4. Weak Magnetic Control of Single Current Regulator Considering Motor Parameters

4.1 Analysis of Weak Magnetic Control Curve for Single Current Regulator

Assuming a three-phase built-in permanent magnet synchronous motor (IPMSM) as an ideal motor, the relationship

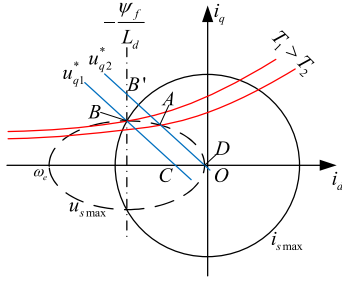


Fig. 4 Current trajectory of d, q axes controlled by constant q-axis voltage flux-weakening control

between the d-axis and q-axis currents can be obtained from Eq. (1):

$$i_q = -\frac{\omega_e L_d}{R_s} i_d + \frac{u_q - \omega_e \psi_f}{R_s} \quad (12)$$

The curve represented by Eq. (12) is also known as the load curve, as shown by the straight line BC in Fig. 4. The coordinates of points B and C are $(-\psi_f/L_d, u_q/R_s)$ and $(-\psi_f/L_d + u_q/\omega_e L_d, 0)$, and in a motor control system using single current regulator weak field control, the final voltage control signal consists of a direct axis current regulator and a fixed quadrature axis voltage signal. When the cross-axis voltage is constant, the steady-state operating point of the motor current will move along the load curve from point C to point B during the loading process, and the load capacity and efficiency of the motor cannot be guaranteed simultaneously.

When the motor is given the quadrature axis voltage at a certain speed, the intersection of the load curve and the horizontal axis is point C, and the intersection of the load curve and the circular symmetry axis of the voltage limit is point B. When the given value of the quadrature axis voltage is u_{q1}^* , the electromagnetic torque required for the motor to output the maximum load capacity is T_1 . Obviously, at this time, the operating point C of the motor at no load is far from the voltage limit ellipse at that speed, and the stator current vector is large, resulting in low operating efficiency of the motor; When the given value of the quadrature axis voltage is u_{q2}^* , the operating point of the motor at no load is point D. This point is on the voltage limit ellipse and is also the minimum stator current that the motor can reach at no load. The voltage utilization rate of the motor during operation is improved compared to when the quadrature axis voltage is u_{q1}^* , but the maximum output electromagnetic torque of the motor is reduced to T_2 . It can be seen that the operating efficiency and load capacity of the motor cannot be simultaneously considered under the traditional weak magnetic field control with fixed AC axis voltage. To simultaneously consider the load capacity and motor efficiency [26], the given method for u_q^* can be determined as:

$$u_q^* = \sqrt{u_{s\max}^2 - u_d^2} \quad (13)$$

The traditional single current regulator weak magnetic

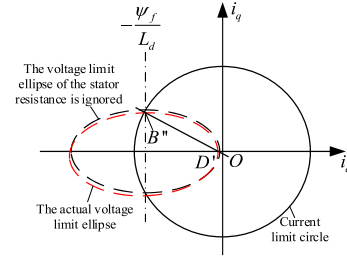


Fig. 5 The voltage limit ellipse and the voltage limit ellipse ignoring R_s

control model usually ignores the influence of motor parameters on the voltage limit vector circle when designing the cross-axis voltage. Among the motor parameters, the stator resistance has the deepest impact on the voltage limit ellipse distortion caused by the motor during high-speed operation [27], [28]. This method maintains the weak magnetic steady-state operating point of IPMSM at the optimal operating point, and the cross-axis voltage command is automatically adjusted according to the motor operating conditions.

4.2 Improved Method for Setting Quadrature Axis Voltage in Single Current Weakening Control

During high-speed operation of the motor, the voltage limiting ellipse expressions that ignore and consider the influence of stator resistance R_s are shown in Eqs. (14) and (15), respectively. From the Eq. (15), the value of R_s will have different effects on the shape and position of the ellipse. It can be seen that R_s appears at multiple positions and is interrelated with other variables. As R_s increases, due to the presence of square and cross terms in the expression, it will affect the coefficients of the ellipse equation, making the ratio of the major and minor axes of the ellipse flatter or slimmer. At the same time, the weights of the terms related to u_d and u_q in the expression will change, causing the ellipse to tilt on the d-q axis and no longer be a standard ellipse.

$$(L_q i_q)^2 + (L_d i_d + \psi_f)^2 \leq (u_{s\max}/\omega_e)^2 \quad (14)$$

$$\begin{aligned} & \left(\frac{u_q \omega_e L_q + R_s u_d - \omega_e^2 L_q \psi_f}{R_s^2 + \omega_e^2 L_d L_q} \right)^2 \\ & + \left(\frac{R_s u_q + R_s^2 \frac{u_d}{\omega_e L_q} - R_s \omega_e \psi_f - u_d}{\omega_e L_q + \frac{R_s^2}{\omega_e L_q} + \omega_e L_d} \right)^2 \\ & \leq (u_{s\max}/\omega_e)^2 \end{aligned} \quad (15)$$

The positions of Eqs. (12) and (14) in the d-q axis coordinates are shown in Fig. 5. It is evident that the long axis and short axis vertices of the voltage limit ellipse considering the influence of stator resistance have shifted towards the lower left of the coordinate axis. The reason is that when the stator resistance increases, the voltage drop on the resistance increases for a given current i_d and i_q , which leads to a decrease in the values of u_d and u_q , causing the limit ellipse to

shift towards the direction of decreasing voltage values, that is, to the lower left. The actual range of voltage limit ellipses available for stable operation of the motor is smaller than that of voltage limit ellipses that ignore stator resistance.

As shown in Fig. 5, ignoring the distortion of the voltage limit vector circle caused by stator resistance and designing a given quadrature axis voltage may guide the steady-state operating point of the motor beyond the voltage limit vector circle, causing the direct quadrature axis current to deviate from tracking and causing the motor control system to lose control. This article proposes a new method for determining u_q^* , which introduces the influence factor of stator resistance R_s to give the quadrature axis voltage, constructs a proportional factor K , and can adjust the degree of change of the given value of the quadrature axis voltage to meet the stability of the motor during high-speed operation. Among them, the command value of the quadrature axis voltage can be given by the following equation:

$$u_q^* = K \sqrt{u_{s\max}^2 - u_d^2} \quad (16)$$

According to Eq. (16), in order to prevent u_q^* from exceeding $u_{s\max}$, $u_d = 0$ is set, and $u_q^* = Ku_{s\max}$, so its upper limit is 1. When the load curve passes through point B, the given value u_q^* of the quadrature axis voltage is $u_{s\max}R_s/\omega_e L_d$, and it still intersects with the d-axis at point C. To ensure both the load capacity and voltage utilization of the motor during operation, the load curve must pass through points B and D simultaneously, and be close to the edge of the voltage limit ellipse. To solve the slope of the BD straight line, an auxiliary resistance coefficient R'_s is introduced to perform auxiliary analysis on the DC axis current plane of the motor operation. By translating the q axis in the d-q axis coordinate system, it can be obtained that:

$$\begin{cases} i'_d = i_d + \frac{\psi_f}{L_d} \\ i'_q = i_q \end{cases} \quad (17)$$

The motor voltage equation also becomes:

$$\begin{cases} u_d = -\omega_e L_q i'_q \\ u_q = \omega_e L_d i'_d \end{cases} \quad (18)$$

The voltage limiting circle equation becomes:

$$(L_q i'_q)^2 + (L_d i'_d)^2 = (u_{s\max}/\omega_e)^2 \quad (19)$$

The load curve equation becomes:

$$i'_q = -\frac{\omega_e L_d}{R_s} i'_d + \frac{u_q}{R_s} \quad (20)$$

At this point, substitute the auxiliary resistance coefficient R'_s into Eq. (20):

$$i'_q = -\frac{\omega_e L_d}{R'_s} i'_d + \frac{u'_q}{R'_s} \quad (21)$$

Combining Eqs. (20) and (21), we can obtain:

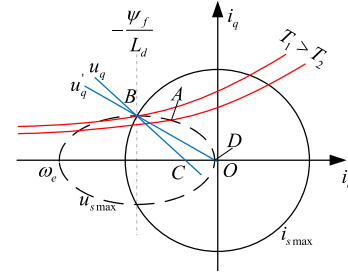


Fig. 6 Running state analysis of PMSM under flux-weakening control

$$u'_q = \frac{R'_s}{R_s} u_q + \omega_e L_d i'_d \left(1 - \frac{R'_s}{R_s}\right) \quad (22)$$

As shown in Fig. 6, the load curve shown in Eq. (12) is called load curve u_q , and the load curve after introducing the auxiliary resistance coefficient is called load curve u'_q . When both the traditional load curve and the load curve with auxiliary resistance coefficient pass through point B, the coordinate of point B is $(0, u_{s\max}/\omega_e L_d)$, while the traditional load curve u_q is $u_{s\max}R_s/\omega_e L_d$. The load curve u'_q after adding auxiliary resistance coefficient is $u'_q \omega_e L_d / u_{s\max}$, with a slope of $-u_{s\max}L_d/u'_q L_q$. The load curve after adding auxiliary resistance coefficient can be obtained as follows:

$$i'_q = -\frac{u_{s\max}L_d}{u'_q L_q} i'_d + \frac{u_{s\max}}{\omega_e L_d} \quad (23)$$

At this point, both the traditional load curve and the load curve with added auxiliary resistance coefficient pass through point B. However, the motor stator current operating point of the load curve with added auxiliary resistance coefficient is further away from the center of the voltage limit ellipse, and the current operating point is closer to the boundary of the voltage limit ellipse. The voltage utilization rate is significantly higher than that of the traditional load curve. When the two overlap

$$u'_q = \frac{u_{s\max}}{\omega_e L_q} R_s \quad (24)$$

So as long as it is satisfied

$$\frac{u_{s\max}}{\omega_e L_q} R_s < u'_q \leq u_{s\max} \quad (25)$$

Adding auxiliary resistance coefficient to load curve u'_q can ensure that the slope is less than load curve u_q , that is, when the motor's carrying capacity is the same, the improved single current regulator weak field control algorithm represented by load curve u'_q has a higher working efficiency than the fixed quadrature axis voltage weak field control algorithm represented by load curve u_q .

Define at this point

$$u'_q = Ku_{s\max} \quad (26)$$

It can be determined that the range of K values is

$$R_s/\omega_e L_q < K \leq 1 \quad (27)$$

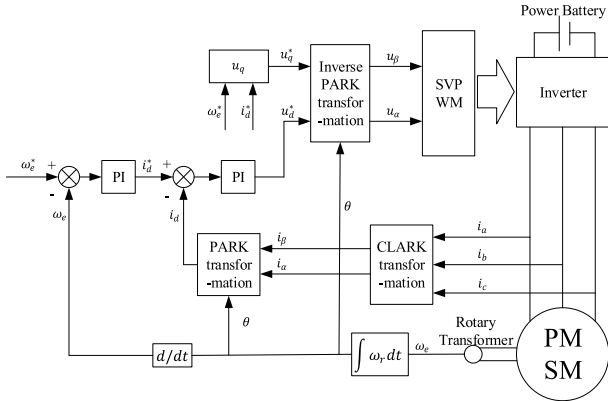


Fig. 7 Improved flux-weakening control system of PMSM

Analysis shows that the larger the K value, the closer the d and q axis voltage command values are to the true voltage limit ellipse boundary, that is, the higher the voltage utilization rate of the inverter output. In the simulation, in order to maintain the voltage synthesis vector of the motor operating in the weak magnetic zone at the maximum output value u_{smax} of the inverter, $K = 1$ is taken. In the experiment, it is necessary to ensure that there is a margin in the stator voltage to keep the back electromotive force of the motor within a controllable range and improve the stability of the motor during operation, $K = 0.95$ is taken.

The improved weak field control system of permanent magnet synchronous motor is shown in Fig. 7.

5. Simulation and Experimental Research

5.1 Simulation Analysis

Based on the above analysis, a simulation model of IPMSM weak magnetic control system was built in Matlab simulation software [29]. Among them, the motor adopts maximum torque current ratio (MTPA) control below the base speed, and weak magnetic control above the base speed. The basic parameters of IPMSM are shown in Table 1.

The simulation duration is 1 second, and the motor is started with a load of 10 N·m to a speed of 3500 r/min; At $t = 0.3$ s, the speed increases to 4000 r/min, and at $t = 0.5$ s, a load of 85 N·m is added. The PI controller parameter setting for the speed loop has a proportional coefficient of 0.963 and an integral coefficient of 6.55. The parameter setting for the current loop PI controller is calculated based on the motor parameters. In both simulation and experiment, parameter identification is carried out online. Due to the different performance and delay of the equipment during the simulation and experiment process, there may be some variations in the experimental results.

By comparing with the basic MBO algorithm and the adaptive monarch butterfly algorithm (ALMBO) proposed in Ref. [26] for mutation reverse learning, the effectiveness of the ACMBO algorithm is verified. The parameters of the three algorithms are set as follows: population size of

Table 1 IPMSM parameter

Parameter	Value
Rated power/kW	60
Bus voltage /V	540
Rated current /A	160
Motor base speed value /(rpm)	3800
Polar logarithm	4
Stator resistance/ Ω	2.50
Flux linkage /Wb	0.1194
d-axis inductance/ μ H	542
q-axis inductance/ μ H	856

Table 2 Parameter identification results of three algorithms

Parameter	MBO	ALMBO	ACMBO
R_s/Ω	2.413	2.532	2.526
deviation/%	3.48	1.28	1.04
ψ_f /Wb	0.1246	0.1178	0.1181
deviation/%	4.35	1.34	1.08
L_d/μ H	498	559	531
deviation/%	8.11	3.13	1.92
L_q/μ H	774	824	846
deviation/%	9.57	3.74	1.16
fitness value	1.61	0.52	0.29

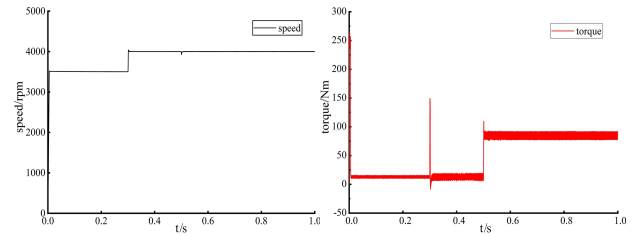


Fig. 8 Improving weak magnetic control of single current regulator

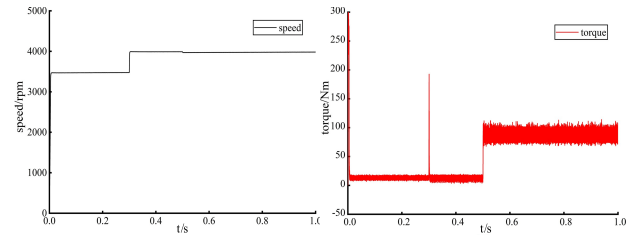


Fig. 9 Traditional fixed axis voltage weak magnetic control

200, maximum number of iteration ξ is 200, $S_{max} = 1$, $BAR = 5/12$. The three algorithms were independently run 30 times, and their average was taken as the experimental result for comparison. The results are shown in Table 2.

According to Table 2, ACMBO can achieve good results in identifying various parameters of the motor, and its convergence speed is faster compared to the other two algorithms. The traditional fixed axis voltage single current weak magnetic control algorithm and the improved single current regulator weak magnetic control strategy proposed in this paper are used for constant speed loading simulation. The simulation waveforms of motor speed and output torque during constant speed loading are shown in Figs. 8 and 9.

The simulation results show that the motor, based on the improved weak magnetic control algorithm, can quickly reach a speed of 3500 r/min during startup, and can quickly

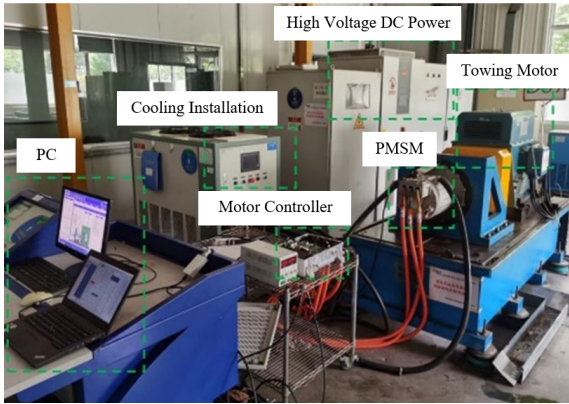


Fig. 10 PMSM drive system experiment platform

stabilize at 4000 r/min when the speed rises within 0.3 s; When a load of 85 N·m is added in 0.5 s, the speed can quickly recover and stabilize at 4000 r/min, and the torque output fluctuation range is less than 3.7%. The control effect is significantly better than the weak magnetic control of traditional fixed axis voltage.

5.2 Experiment Analysis

The experimental platform for the permanent magnet synchronous motor transmission system consists of two parts: the high-voltage drive part and the low-voltage control part, as shown in Fig. 10.

The high-voltage drive part mainly includes high-voltage DC power supply, power analyzer, PMSM and its drive circuit, load dynamometer, and heat dissipation system, and the low-voltage control part includes low-voltage power supply (24 V), CAN communication, and upper computer.

During the experiment, the initial speed of the motor was 3500 r/min, and the initial load torque was 10 N·m. The voltage signal was measured using the method proposed in Ref. [30]. We use CAN communication to send the collected voltage, current, and speed signals to the upper PC, and utilize the powerful computing capabilities of the upper PC for parameter identification and calculation. To ensure the accuracy of the experimental results and the consistency of the simulation, the parameter settings of the three identification algorithms are the same as those in the simulation experiment. The experimental results are shown in the following figures, where Fig. 11 is the average fitness curve and Fig. 12 is the identification curve of each parameter. The comparison of parameter identification results among the three algorithms is shown in Table 3.

From Figs. 11 and 12, it can be seen that all three types of Monarch Butterfly Optimization can converge to the actual values of motor parameters after a certain number of iterations. Compared with the basic MBO algorithm, the two improved Monarch Butterfly Optimization algorithms have more accurate identification results, indicating that both algorithms have strong global optimization capabilities.

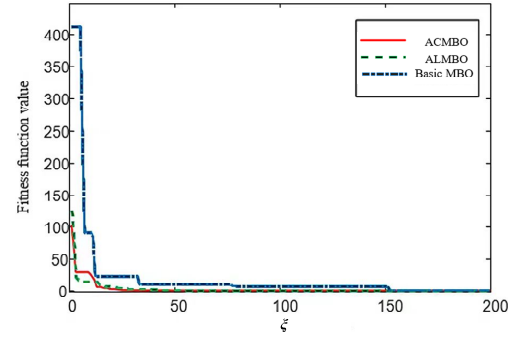


Fig. 11 Curve of average fitness value

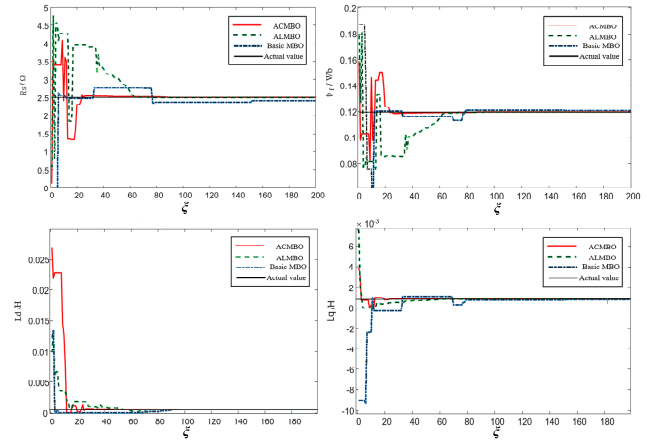


Fig. 12 Identification curves for stator(R_s), flux linkage(ψ_f), d-axis inductance(L_d), and q-axis inductance(L_q)

Table 3 Comparison of parameter identification experimental results of the three algorithms

Parameter	MBO	ALMBO	ACMBO
R_s/Ω	2.776	2.613	2.587
deviation/%	11.04	4.52	3.48
ψ_f/Wb	0.1256	0.1233	0.1221
deviation/%	9.31	3.28	2.31
$L_d/\mu\text{H}$	481	577	553
deviation/%	11.33	6.52	2.12
$L_q/\mu\text{H}$	766	812	829
deviation/%	10.54	5.33	3.27
fitness value	2.58	1.32	0.74

Among them, the ACMBO algorithm proposed in this paper has a significantly faster convergence speed, and can converge to the vicinity of the optimal value after 30 iterations. By analyzing the initial and actual values of each parameter identification curve, and calculating the degree of approximation between the two values, it can be concluded that 42% of the previous parameter identification results were reflected in the initial population, indicating that the algorithm fully utilized the previous identification results.

When the motor adopts an improved single current regulator weak field control, after the motor speed stabilizes, continue to increase the motor speed to 4000 r/min. At this time, the motor works in the weak field area, and the load torque gradually increases to 85 N·m. The experimental re-

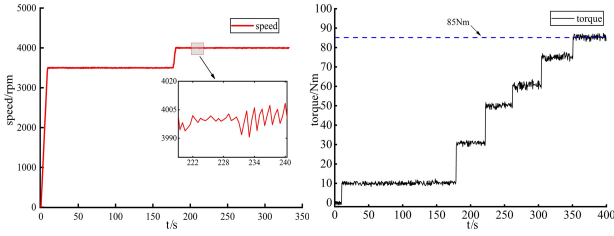


Fig. 13 Improving weak magnetic control of single current regulator

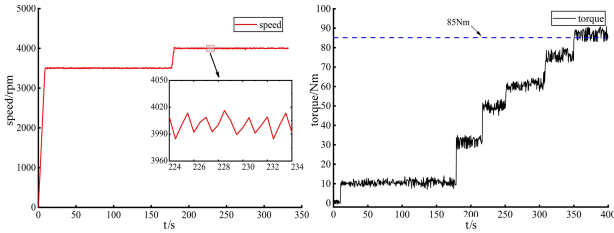


Fig. 14 Traditional fixed axis voltage weak magnetic control

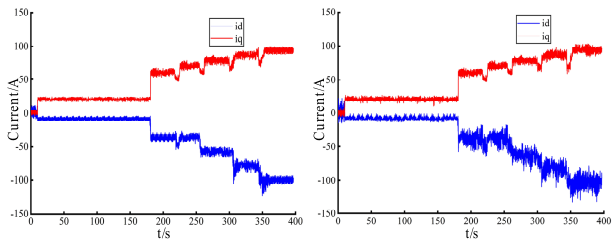


Fig. 15 Experimental curves of d and q-axis current

sults are shown in Figs. 13 and 14.

At 3500 r/min and 4000 r/min, the control mode switching of the motor is smooth, and the speed fluctuation and speed response time are small. As shown in Fig. 14, the motor has a short torque response time, can accurately output 85 N·m of torque, and the error range is within 3.5%. The trend of changes in experimental and simulation results is basically consistent, with some segments showing deviations due to experimental environment and measurement errors, all within a reasonable range. The experimental results show that improving the weak magnetic strategy can make PMSM operate more smoothly in the high-speed region and improve the smoothness of operation under high-speed and high torque conditions. The d and q-axis current curves recorded by the two methods in the experiment are shown in Fig. 15.

It can be clearly seen that when using the improved single current regulator weak field control, the d and q-axis current curves of the motor fluctuate less, which proves the improvement of motor operation stability by this control method; Among them, under the same load conditions, there is no significant change in the magnitude of the q-axis current, but the amplitude of the d-axis current decreases compared to the weak magnetic control of traditional single current regulators, that is, the amplitude of the stator current

of the motor decreases.

According to the analysis of Figs. 4, 5, and 6, it can be concluded that the operating point of traditional fixed axis voltage weak magnetic control will gradually move towards the weak magnetic region. As the torque increases, in order to meet the torque demand, the q-axis current will continue to increase. At the same time, in order to prevent the voltage from exceeding the limit, the d-axis current will be adjusted in the demagnetization direction to weaken the magnetic field. However, this adjustment will cause the motor operating point to enter an unstable region, resulting in a decrease or fluctuation in torque. For improving the weak magnetic control method of a single current regulator, the d-q axis current coordinated changes during high torque, and the operating point will smoothly move towards the weak magnetic region, which can avoid weakening torque fluctuations during high torque due to excessive demagnetization. The stator current operating point of the improved method's high torque load curve motor is further away from the center of the voltage limit ellipse, that is, the intersection point of the load curve and the horizontal axis moves to the right, and the current operating point is closer to the voltage limit ellipse boundary. However, the stator current operating point of the traditional method is relatively far away from the voltage limit ellipse boundary, and the voltage utilization rate is significantly lower than the improved method's load curve. This is because traditional methods have not fully considered the influence of stator resistance. Under high torque conditions, the voltage drop across the stator resistance becomes significant, which results in the actual input voltage to the motor not matching the expected value, thereby affecting the operational stability of the motor and causing torque fluctuations.

6. Conclusion

This paper proposes an improved Monarch Butterfly Optimization based on information transmission and chaotic mapping for online parameter identification of permanent magnet synchronous motors for vehicles, and solves the problem of under rank in the identification equation by constructing a steady-state full rank model of the motor. A weak magnetic control method for improving the single current regulator of permanent magnet synchronous motor has been designed, which stabilizes the steady-state operating point of the motor within the voltage limit ellipse and avoids the controller overvoltage caused by the easy detachment of the vehicle from the voltage limit ellipse under high speed and high torque conditions. Simulation and experimental studies have shown that the control method proposed in this paper reduces the identification errors of stator resistance, permanent magnet net flux, d-axis inductance, and q-axis inductance to 3.48%, 2.31%, 2.12%, and 3.27%, respectively. When the load is 85 N·m, the error range is less than 3.5%. This indicates that the improved single current regulator weak field control is feasible and effective. Due to the influence of experimental environment and measurement er-

rors, there are deviations in some aspects. Further research can be conducted using more precise automation equipment.

References

- [1] X. Wang, J. Diao, L. Wang, X. Zhao, and S. Yu, "Eddy current loss analysis of permanent magnets of ISG permanent magnet motor for electric vehicle under weak magnetic conditions," *Journal of Mechanical Engineering*, vol.56, no.12, pp.155–164, 2020.
- [2] H. Liu, and S. Li, "Speed control for PMSM servo system using predictive functional control and extended state observer," *IEEE Trans. Ind. Electron.*, vol.59, no.2, pp.1171–1183, 2012.
- [3] S. Li, and Z. Liu, "Adaptive speed control for permanent-magnet synchronous motor system with variations of load inertia," *IEEE Trans. Ind. Electron.*, vol.56, no.8, pp.3050–3059, 2009.
- [4] X. Zhang, J. Shi, Y. Wang, and Y. Liu, "MTPA control of precise parameters for interior permanent magnet synchronous motor," *Electrical Measurement & Instrumentation*, vol.60, no.10, pp.124–128, 2023.
- [5] S. Chi, L. Xu, and Z. Zhang, "Efficiency-optimized flux-weakening control of PMSM incorporating speed regulation," 2007 IEEE Power Electronic Specialists Conference, Orlando, FL, USA, pp.1627–1633, 2007.
- [6] Z. Zhang, C. Wang, M. Zhou, and X. You, "Flux-Weakening in PMSM Drives: Analysis of Voltage Angle Control and the Single Current Controller Design," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol.7, no.1, pp.437–445, 2019.
- [7] Y. Zhang, S. Guo, Z. Wu, T. Chen, and N. Liu, "Parameter identification of permanent magnet synchronous motor based on multi-innovation limited memory method," *Science Technology and Engineering*, vol.21, no.13, pp.5353–5359, 2021.
- [8] H. Zheng, J. Hao, M. Zha, Z. Liu, and W. Kong, "Sensorless Control and Inductance Parameter Identification of PMSM Based on Two-Orientation High-Frequency Square Wave Injection," 2023 IEEE 6th International Electrical and Energy Conference (CIEEC), Hefei, China, pp.585–590, 2023.
- [9] A.G. Bartsch, G.H. Negri, M.S.M. Cavalca, J. de Oliveira and A. Nied, "Cost function tuning methodology for FCSMPC applied to PMSM drives," 2017 Brazilian Power Electronics Conference (COBEP), Juiz de Fora, Brazil, pp.1–6, 2017.
- [10] I.M. Alsofyani, and K.-B. Lee, "A unidirectional voltage vector pre-selection strategy for optimizing model predictive torque control with discrete space vector modulation of IPMSM," *IEEE Trans. Ind. Electron.*, vol.69, no.12, pp.12305–12315, Dec. 2022.
- [11] L. Wang, S. Zhang, C. Zhang, and Y. Zhou, "An Improved Deadbeat Predictive Current Control Based on Parameter Identification for PMSM," *IEEE Trans. Transport. Electrification*, vol.10, no.2, pp.2740–2753, June 2024.
- [12] J. Wu, C. Xie, Z. Guo, and Y. Wang, "An optimized sliding-mode observer control algorithm for vehicle sensor-less PMSM," *Journal of Engineering*, vol.2023, no.3, pp.1–9, 2023.
- [13] W. Zhang, C. Liu, C. Lian, J. Liu, and Z. Mai, "Optimization of SVPWM Algorithm used in PMSM DTC," 2023 3rd International Conference on Electrical Engineering and Mechatronics Technology (ICEEMT), Nanjing, China, pp.492–495, 2023.
- [14] P.Q. Khanh, N.T. Dat, and H.P.H. Anh, "Optimal Fuzzy-PI PMSM Speed Control Using Evolutionary DE Algorithm Implemented on DSP Controller," 2023 International Conference on System Science and Engineering (ICSSE), Ho Chi Minh, Vietnam, pp.206–211, 2023.
- [15] Y.L. Karnavas, I.D. Chasiotis, M.S.C. Pechlivanidou, E.K. Karamanis, and A.G. Kladas, "Azimuth Thruster PMSM Optimization using Symbiotic Organisms Search Algorithm," 2020 International Conference on Electrical Machines (ICEM), Gothenburg, Sweden, pp.2231–2237, 2020.
- [16] X. Yun, X. Zhou, F. He, and D. Wang, "PMSM dual vector model predictive control based on voltage error judgment," *Power Electronics*, vol.53, no.1, pp.11–13+25, 2019.
- [17] C. Qi, and Z. Song, "Controlling of permanent-magnet synchronous machine based on particle swarm optimization fuzzy logical controller," *Proc. CSEE*, vol.26, no.17, pp.158–162, 2006.
- [18] X. Liu, Y. Pan, Y. Zhu, H. Han, and L. Ji, "Decoupling Control of Permanent Magnet Synchronous Motor Based on Parameter Identification of Fuzzy Least Square Method," *Progress In Electromagnetics Research M*, vol.103, pp.49–60, 2021.
- [19] C. Zhong, and Y. Lin, "Model reference adaptive control (MRAC) based parameter identification applied to surface mounted permanent magnet synchronous motor," *International Journal of Electronics*, vol.104, no.11, pp.1854–1873, 2017.
- [20] Z. Wang, M. Zhou, B. Xie, L. Bai, and C. Wang, "A control strategy of PMSM based on D-axis current regulator under square wave," *Proc. CSEE*, vol.42, no.3, pp.1154–1163, 2022.
- [21] Y. Liang, "Research on current control strategy in deep field weakening process of permanent magnet synchronous motor," M.D. Dissertation, Jiangsu University of Science and Technology, 2022.
- [22] G.-G. Wang, S. Deb, and Z. Cui, "Monarch butterfly optimization," *Neural Computing & Applications*, vol.31, no.7, pp.1995–2014, 2019.
- [23] J. Li, S. Yang, Z. Xie, and X. Zhang, "Online parameter identification of permanent magnet synchronous motor based on effective information iterative fast particle swarm optimization algorithm," *Transactions of China Electrotechnical Society*, vol.37, no.18, pp.4604–4613, 2022.
- [24] O.M. Dorgham, M. Alweshah, M.H. Ryalat, J. Alshaer, M. Khader, and S. Alkhalailah, "Monarch butterfly optimization algorithm for computed tomography image segmentation," *Multimedia Tools and Applications*, vol.80, no.20, pp.30057–30090, 2021.
- [25] X. Fu, H. Gu, G. Chen, J. Zou, and J. Zhang, "Permanent magnet synchronous motors parameters identification based on cauchy mutation particle swarm optimization," *Transactions of China Electrotechnical Society*, vol.29, no.5, pp.127–131, 2014.
- [26] X. Fang, T. Hu, F. Lin, and Z. Yang, "Single current regulator flux-weakening control of PMSM based on current cross-coupling effect," *Transactions of China Electrotechnical Society*, vol.30, no.2, pp.140–147, 2015.
- [27] X. Shi, Y. Yi, and W. Shi, "Optimization of single current regulator flux-weakening control strategy for permanent magnet synchronous motor," *Control Engineering of China*, vol.28, no.2, pp.327–334, 2021.
- [28] C. Liu, "Research on single current regulator flux-weakening control system of permanent magnet synchronous motor," M.D. Dissertation, Hebei University of Technology, 2022.
- [29] L. Yuan, ed., *Modern permanent magnet synchronous motor control principle and MATLAB simulation*, Beihang University Press, Beijing, 2016.
- [30] C. Wu, Y. Zhao, and M. Sun, "Multiparameter online identification for permanent magnet synchronous machines using voltage measurements," *Proc. CSEE*, vol.40, no.13, pp.4329–4339, 2020.



Yihao Wang was born in China. In 2022, he graduated from Zhongyuan University of Technology in Zhengzhou, China, majoring in vehicle engineering, with a Bachelor's degree in Engineering. He is a master student at the School of Vehicle and Transportation Engineering, Henan University of Science and Technology. The main research directions are new energy vehicles.



Jianguo Xi is a Master & corresponding author. Master's supervisor and senior laboratory technician from Henan University of Science and Technology. He obtained a Master's degree in Vehicle Application Engineering from Chang'an University in 2008. The main research directions are new energy vehicle technology and intelligent connected vehicle technology.



Chengwei Xie was born in China. In 2022, he graduated from Henan University of Science and Technology in Luoyang, China, majoring in vehicle engineering, with a Master's degree in Engineering. He is a doctor student at the School of Vehicle and Transportation Engineering, Henan University of Science and Technology. The main research directions are intelligent connected vehicle technology.